PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Metabolic risk factors of non-communicable diseases in Ethiopia: a systematic review and meta-analysis
AUTHORS	Alamnia, Tilahun Tewabe; Tesfaye, Wubshet; Abrha, Solomon; Kelly, Matthew

VERSION 1 – REVIEW

REVIEWER	Bezerra Bueno, Nassib Universidade Federal de Alagoas
REVIEW RETURNED	08-Mar-2021

GENERAL COMMENTS	In my opinion, the authors satisfactorily answered all my raised concerns. I have no further comments regarding this manuscript.

REVIEWER	Morgan, Rebecca McMaster University, Department of Clinical Epidemiology & Biostatistics
REVIEW RETURNED	12-Aug-2021

GENERAL COMMENTS	Colleague assisting with review of the manuscript: Ifeoluwa Babatunde The author present the results of a systematic review and meta-analysis of select non-communicable disease prevalence estimates. In addition, they assess the presence of an association between select NCD outcomes and sex and age. Readers would benefit from additional clarity of the research objective and how that is organized based on the population, exposure and outcome framework proposed by authors. Has this been used elsewhere in the literature? It seems as though this may not be an appropriate fit for the question of: pooled prevalence, where only the primary studies reporting on adults living in Ethiopia with the outcomes of interest are eligible. From that body of literature, it then seems like further exploration of the association with sex and age are wanted. How do the exposures of “any type of variables” contribute to these two main research questions? Only PubMed search strategy is provided. Author should consider including all search strategies in the appendix. Additional information and rationale is needed to justify the subgroups analyzed. Is it appropriate to include studies conducted before 2010 in the review and analyses?

	Authors compare the prevalence findings in this current review to the 2014 report by the WHO, indicating a decrease in the pooled prevalence of hypertension in the Ethiopian population when compared to the older WHO report (Discussion page 15, line 15 – 22). The authors state that this decrease is due to certain lifestyle changes observed within the population in the last 5 years. It might be inappropriate to include prevalence studies conducted more than 5 years ago into their pooled analysis as proof of a recent decrease in the prevalence of hypertension. Prevalence estimates are provided; however, they do not seem to take the results from the risk of bias assessment into account. Should/were any studies excluded from the analysis because of concerns for bias? Did any studies adjust for critical confounders? The substantial heterogeneity seen in the analysis is left largely unexplained by the authors. How do the authors believe the heterogeneity affects the strength and quality of their evidence? Assessments of the association of sex and age were conducted; however, it's unclear whether only the studies that provided significant findings were summarized in the narrative or if all studies reporting on this association provided significant findings. Additional justification is needed for these methods and why they are not selectively reporting on only studies reporting a significant association. How is risk of bias integrated into the assessment of the association? It would be helpful to take these analyses further to feedback into how much certainty we have about the estimates of NCD. If sex and age are associated, should they be taken into account in the model? The authors mention that their pooled analysis produces "strong evidence to inform decision making." However, the authors do not provide the framework (e.g. GRADE) through which they appraised the certainty or quality of the evidence for this research question. A discussion of these results using the domains in GRADE's certainty of evidence framework (Risk of bias, Indirectness, Inconsistency, Imprecision) would be useful in strengthening the recommendations made by the authors in the discussion section. Currently, it is unclear how the strength of the evidence is driving the recommendations made by the author. In the discussion, the authors compare the prevalence rates with previously published rates; however, this discussion would greatly benefit from a discussion of the quality of previously published rates and how certain the authors are that the rates presented can be compared. The conclusions may be overstated given the limited discussion of the quality of the underlying evidence. Further assessment of the risk of bias findings and reasons for substantial heterogeneity are needed to support the conclusions in the paper. [Suggestion: Moberg J, Oxman AD, Rosenbaum S, et al. The GRADE Evidence to Decision (EtD) framework for health system and public health decisions. Health Res Policy Syst. 2018;16(1):45. Published 2018 May 29. doi:10.1186/s12961-018-0320-2]

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Nassib Bezerra Bueno, Universidade Federal de Alagoas
Comments to the Author:
In my opinion, the authors satisfactorily answered all my raised concerns. I have no further comments regarding this manuscript.
Response: Thank you, Dr. Nassib Bezerra Bueno, for your time and expertise in reviewing our manuscript and your previous valuable comments that made us to significantly improve this work.
Reviewer: 2
Dr. Rebecca Morgan, McMaster University
Comments to the Author:
Colleague assisting with review of the manuscript: Ifeoluwa Babatunde
The author presents the results of a systematic review and meta-analysis of select non-communicable disease prevalence estimates. In addition, they assess the presence of an association between select NCD outcomes and sex and age.
Response: Thank you, Dr. Rebecca Morgan, and your colleague (Ifeoluwa Babatunde), for your time and valuable comments, all of which have helped us to improve our work. We have responded to your comments below and made appropriate changes in the manuscript as required.
Readers would benefit from additional clarity of the research objective and how that is organized based on the population, exposure and outcome framework proposed by authors. Has this been used elsewhere in the literature? It seems as though this may not be an appropriate fit for the question of pooled prevalence, where only the primary studies reporting on adults living in Ethiopia with the outcomes of interest are eligible. From that body of literature, it then seems like further exploration of the association with sex and age are wanted. How do the exposures of “any type of variables” contribute to these two main research questions?
Response: Thanks for your suggestions making our research objectives clearer using ‘population, exposure and outcome’ parameters. We have now updated it based on the comments (page 4: line 20-30).
Only PubMed search strategy is provided. Author should consider including all search strategies in the appendix.
Response: Thanks to this, we have now attached the search techniques used and results obtained in each database as an additional file (supplementary table 1).
Additional information and rationale are needed to justify the subgroups analyzed.
Response: Thank you, we have elaborated this in the methods sections (page 9: line 21-32, and page 10: line 1-7).
Is it appropriate to include studies conducted before 2010 in the review and analyses?
Response: In the past few years, Ethiopia has undergone lots of changes in various aspects including social, political, economic areas and the country is on the way to epidemiological transitions wherein non-communicable diseases are becoming the leading causes of deaths. Despite these, limited number of publications were available on the prevalence of metabolic risk factors for non-communicable diseases. Because of this we did not limit the search parameters. We thought including the whole range of literature would give a comprehensive evidence of metabolic risk factors and their change over time among adults to inform policy priority for action. In addition, we performed subgroup analysis to show differences in prevalence between year ranges as shown in Table 1.
Authors compare the prevalence findings in this current review to the 2014 report by the WHO, indicating a decrease in the pooled prevalence of hypertension in the Ethiopian population when compared to the older WHO report (Discussion page 15, line 15 – 22). The authors state that this decrease is due to certain lifestyle changes observed within the population in the last 5 years. It might be inappropriate to include prevalence studies conducted more than 5 years ago into their pooled

analysis as proof of a recent decrease in the prevalence of hypertension.
Response: Thanks for this comment, appreciated. We have now updated this section to highlight the contribution of metabolic risk factors for non-communicable and variations by age, gender, and regions (page 16: line 23-28).
Prevalence estimates are provided; however, they do not seem to take the results from the risk of bias assessment into account. Should/were any studies excluded from the analysis because of concerns for bias? Did any studies adjust for critical confounders?
Response: The risk of bias of studies was first assessed, and most (89%) them had low risk of bias and the remaining (11%) had moderate level of risk of bias. No study was found to have a high risk of bias. Besides, in our confounding assessment we did not identify potential sources of bias. Therefore, we have included all of the studies in the analysis, no study was excluded due to high-risk bias (page 11: line 23-27). The risk bias score of each study was included in the supplementary table 2).
The substantial heterogeneity seen in the analysis is left largely unexplained by the authors. How do the authors believe the heterogeneity affects the strength and quality of their evidence?
Response: There was a substantial heterogeneity among the included studies that might limit the quality of our evidence. In the current version, we have discussed the effect of heterogeneity in the certainty of our evidence throughout the document (starting from page 12 starting line 5).
Assessments of the association of sex and age were conducted; however, it's unclear whether only the studies that provided significant findings were summarized in the narrative or if all studies reporting on this association provided significant findings. Additional justification is needed for these methods and why they are not selectively reporting on only studies reporting a significant association. How is risk of bias integrated into the assessment of the association? It would be helpful to take these analyses further to feedback into how much certainty we have about the estimates of NCD. If sex and age are associated, should they be taken into account in the model?
Response: Thanks for this, we have considered studies only reporting significant associations between the age and sex and metabolic factors. This is because 1) studies included in this review provided odds ratio and confidence intervals only when significant associations with sex and age was observed (page 9: line 7-12). Also, 2) as we mentioned earlier, we did not have studies with high risk of bias and thus did not exclude any study from estimates due to risk of bias. The certainty of the evidence is also discussed in the current version. Besides, in the current version, we indicated the certainty evidence (low) and its effect in the interpretation of our findings (Page 11: line 27-29)
The authors mention that their pooled analysis produces "strong evidence to inform decision making." However, the authors do not provide the framework (e.g., GRADE) through which they appraised the certainty or quality of the evidence for this research question. A discussion of these results using the domains in GRADE's certainty of evidence framework (Risk of bias, Indirectness, Inconsistency, Imprecision) would be useful in strengthening the recommendations made by the authors in the discussion section. Currently, it is unclear how the strength of the evidence is driving the recommendations made by the author.
Response: Thanks for suggesting this, in the current version, we have incorporated and discussed the certainty of the evidence of our findings using the GRADE approach (Page 11: starting line 27)
In the discussion, the authors compare the prevalence rates with previously published rates; however, this discussion would greatly benefit from a discussion of the quality of previously published rates and how certain the authors are that the rates presented can be compared.
Response: We tried to incorporate quality score differences among studies in the discussion section, given limited report on this regard in the literatures, previous evidence.
The conclusions may be overstated given the limited discussion of the quality of the underlying evidence. Further assessment of the risk of bias findings and reasons for substantial heterogeneity are needed to support the conclusions in the paper.
Response: We have updated the conclusion based on the comment to include risk of bias and heterogeneity (page 22: line 6-9).
[Suggestion: Moberg J, Oxman AD, Rosenbaum S, et al. The GRADE Evidence to Decision (EtD) framework for health system and public health decisions. Health Res Policy Syst. 2018;16(1):45.

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Response: Thank you for the suggested reading which really helped to appreciate the GRADE guidelines.